

Victoria Xu

vx18@ic.ac.uk

OVERVIEW

Final year structural biology PhD student at Imperial College London.

EDUCATION

- **Life Sciences (PhD) - Imperial College London** – London, UK
Oct 2022 – Mar 2026
Funding: Imperial College London Life Sciences and AstraZeneca
Supervisors: Sarah L. Rouse, Doryen Bubeck, Matthew Wake (AstraZeneca)
 - Project: *Countdown to Package: Structural Study of Replication Protein 40 in the Adeno-associated Virus Packaging Machinery*
 - Structural biology project carried out in pharmaceutical and academic labs as well as national cryo-EM facility (eBIC)
- **Molecular and Cellular Biosciences (MRes) - Imperial College London** – London, UK
Oct 2021 – Oct 2022
 - Project 1: *Characterising the Interaction of OPA1 Long- and Short-isoforms with Cardiolipin-containing Membranes*
 - Project 2: *Role of CD59 in Cellular Defence of Complement Attack Proteins*
- **Biochemistry (BSc) - Imperial College London** – London, UK
Oct 2018 – Oct 2021

PUBLICATION AND GRANTS

- **Xu, V.**, McInnes, A., Wake, M., Acebrón-García-de-Eulate, M., Barritt, J. D., Bubeck, D., Rouse, S. L., ***Structural basis for Rep-mediated adeno-associated virus packaging. Cell Reports***
- UKRI Impact Acceleration Account (IAA) award – Named Investigator.

CONFERENCES

- **Session chair:**
 - Biophysical Society (BPS) annual meeting protein-nucleic acid platform I. Los Angeles, CA, USA. 2025
- **Talks:**
 - BPS annual meeting platform speaker. Los Angeles, CA, USA. 2025 ([https://www.cell.com/biophysj/abstract/S0006-3495\(24\)00992-5](https://www.cell.com/biophysj/abstract/S0006-3495(24)00992-5))
 - Flatiron institute Structural and Molecular Biophysics (SMBp) seminar speaker. New York, NY, USA. 2025 (<https://indico.flatironinstitute.org/event/4067/>)
 - Imperial Centre for Structural Biology open day speaker. London, UK. 2025
- **Poster:**
 - British Crystallographic Association winter meeting. Oxford, UK. 2023
 - Biophysical society (BPS) annual meeting. Los Angeles, CA, USA. 2025
 - American Society for Gene and Cell Therapy (ASGCT). New Orleans, LA, USA. 2025 (<https://www.sciencedirect.com/science/article/pii/S1525001625003648>)
- **Participant:**
 - London Consortium for High-Resolution Cryo-Electron Microscopy (LonCEM) symposium. London, UK. 2023
 - The Collaborative Computational Project for electron cryo-microscopy (CCP-EM) Spring symposium. Nottingham, UK. 2023, 2024 and 2025.

AWARDS

- Eurofins Foundation Postgraduate Prize for MRes Molecular and Cellular Biosciences (Top student)
- Imperial Department of Life Sciences Travel Award

EXPERIENCE

- **Experimental shifts – Diamond Synchrotron /Electron Bio-Imaging Centre (eBIC) – Oxford, UK**

During PhD

- Grid freezing with Chameleon (SPT) high-speed grid vitrification robot
- Krios microscope operation
- Completed eBIC SPA training

- **Research Assistant Intern - Academia Sinica – Taipei, Taiwan**

Jul 2020 – Nov 2020

Supervisor: Cheng-Hsun, Ho

- Project: Functional Study of Nitrate transporter 1/ Peptide Transporter Family (NPF) in Arabidopsis Thaliana via FRET Biosensor Development and Screening

KEY SKILLS

- **Wet lab:** Cloning, protein expression and purification in E.coli, mammalian cell culture, virus purification, x-ray crystallography, cryo-electron microscopy SPA, ELISA, mass photometry, confocal microscopy
- **Dry lab:** Molecular dynamics simulation (CHARMM-GUI, GROMACS), RELION, Scipion and cryoSPARC image processing, model building (ISOLDE, Coot, phoenix), deposition (PDB, EMDB, EMPIAR).
- **Mentoring:** Supervised Structural Biology MRes student over 10-month project and Biochemistry undergraduate over 3-month project.